Compact Quantum Circuits for Martinican Creole LID: A Typologically Informed Proof-of-Concept

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Abstract. We present the first Quantum-Natural-Language-Processing (QNLP) proof of concept for token-level language identification in Martinican Creole, a low-resource language with substantial lexical overlap with French. Our approach compares: (1) a character-bigram baseline, (2) a classical logistic regression model trained on three interpretable intra-token features (entropy, vowel ratio, syllable complexity), and (3) a compact three-qubit variational circuit embedding the same features in product-state and entangled configurations.

On token-level macro-F_1 and accuracy, the classical model performs slightly better than the entangled variant, reflecting its robustness on high-frequency items where individual features are most discriminative. However, a type-level analysis shows a different pattern: the entangled embedding reduces the number of distinct misclassified types and uniquely corrects a sizable subset that both the classical and product-state models miss. The rescued set is heterogeneous (i.e., mixing frequent Creole items and proper names with cross-lingual orthography), and some gains are dataset-contingent serendipitous effects, since the models are context-blind and rely only on intra-token features.

Our findings demonstrate that even minimal quantum embeddings can capture non-linear interactions between interpretable features in ways that simple linear models and product mappings cannot. The results highlight the potential of hardware-efficient QNLP for fine-grained, low-resource language identification, especially in settings where context is unavailable but token-internal cues are strong. We conclude with a roadmap for extending this work to richer feature sets, context-aware hybrid architectures, and cross-lingual transfer to other Creole and closely related language pairs.

Keywords: QNLP · Martinican Creole · Token-level Language Identification · Low-Resource Language Modeling

1 Introduction

Automatic language identification for Creole text at the token level presents a uniquely challenging problem, especially when discriminating between a Creole

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language and its lexifier. In varieties such as Martinican Creole, constant contact with French gives rise to pervasive code-switching and mixing—both between sentences (intersentential) and within a single clause (intrasentential) [7,8]. While sentence-level language identification can exploit large context windows and strong distributional cues, intrasentential token-level identification must decide on very short strings, often a single word, clitic, or affix whose orthographic and phonotactic properties can overlap entirely across both languages.

To isolate and amplify the subtle structural signals that distinguish Creole from French at this fine granularity, we present the first QNLP proof of concept for token-level Creole identification. By deliberately restricting ourselves to token-internal features, we keep our quantum circuits within the size limits of today's noisy intermediate-scale devices and can pinpoint exactly which interpretable features benefit from quantum interference and entanglement. Each token is mapped to a compact three–dimensional feature vector consisting of: (1) Shannon entropy over the character sequence, (2) vowel-to-consonant ratio, and (3) a simple syllable-complexity measure.

We compare three end-to-end pipelines:

1. A purely classical structural-feature model in which the three–dimensional feature vector is passed into a logistic-regression classifier (with z-score scaling).
2. A product-state quantum pipeline in which the features are embedded into a 3–qubit variational circuit without entanglement, followed by a logistic-regression head.
3. An entangled quantum pipeline in which the same embedding is passed through a 3–qubit variational circuit with Ising entanglers, also decoded with logistic regression.

Linguistic typology and information theory both motivate our choice of features. Creoles typically favor simple (CV) syllable structure without complex clusters [9], and exhibit lower symbol-sequence entropy than typologically related lexifiers [2,10]. Likewise, Creoles cluster around roughly 20 consonants but only 6–8 basic vowels, whereas French employs a similarly sized consonant inventory alongside a much richer vocalic system [4,5]. These cross-linguistic patterns directly inform our entropy, C/V-ratio, and syllable-complexity measures.

In this paper we make three contributions:

1. We establish a robust, interpretable classical baseline for token-level Creole identification using three compact, linguistically motivated features.
2. We perform a systematic QNLP comparison of product-state and entangled 3–qubit variational circuits as the first token-level benchmark for Martinican Creole identification.
3. We provide a comprehensive evaluation (precision, recall, F_1, confusion matrices) and an in-depth type-level, class-conditioned error overlap analysis, highlighting both shared and model-specific misclassifications.

Our overlap analysis shows that the product-state model makes nearly all of the errors made by the classical model, while introducing many more of its own, whereas the entangled model substantially reduces the total number of unique error types. Crucially, entanglement "rescues" a large set of errors shared by the other two models, including frequent, monosyllabic Creole tokens, even though it introduces a small set of unique errors on rare French borrowings. This pattern underscores entanglement's power to capture subtle joint-feature interactions in a compact circuit, while pointing to morphological or contextual augmentations to curb its remaining misclassification.

2 Background

2.1 Token-Level Language Identification

Fine-grained, token-level language identification (LID) is markedly more challenging than document- or sentence-level LID due to extremely short contexts (often 1–7 characters), orthographic overlap between varieties, and severe class imbalance (in our case, far more Creole tokens than French borrowings). Early work on token-level LID in code-switched settings relied on lexicons, character n-gram log-linear models, and simple sequence models (CRFs), achieving F_1 scores in the 60–80% range but suffering high confusion on closely related or shared-orthography tokens.

Molina et al. [6] provide a comprehensive overview of the 2016 shared task on language identification in code-switched data. In their summary of the 2016 evaluation, the best-performing Spanish–English system achieves a token-level F_1 of 0.973, whereas on the typologically closer pair of Modern Standard Arabic vs. Dialectal Arabic the top token-level F_1 falls to 0.876. Neural architectures that learn character- and contextualized word-level embeddings show improvements on social-media text, yet they still underperform on extremely short or out-of-domain tokens [3].

Our work departs from these standard n-gram and lexicon-heavy approaches by systematically evaluating three progressively richer classifiers on a 9k-token Martinican Creole test set, all using the same compact, interpretable token-internal features: entropy, vowel ratio, and syllable complexity. First, a character-bigram baseline model classifies each token via a softmax over their log-likelihoods. Second, we establish a classical logistic-regression baseline trained on these features (with z-score scaling). Third, we embed the same features into a 3-qubit variational circuit in a product-state configuration, and then introduce entanglement via interleaved CNOT gates to capture higher-order joint-feature correlations.

Our type-level, class-conditioned error overlap analysis reveals a clear hierarchy: the product-state model reproduces nearly all errors made by the classical model while adding its own, whereas the entangled model substantially reduces the total number of unique error types and "rescues" many frequent, monosyllabic Creole tokens misclassified by both baselines. These gains come alongside

a small set of unique entangled-only errors, mostly rare French borrowings, highlighting both the promise and the current limits of compact entangled circuits for this task.

2.2 Structural Token Features

Creole syllable structure is uniformly simple, a fact we capture in our *syllable_complexity* feature. Typological surveys show that most Martinican Creole syllables conform to CV or VC templates, systematically shedding complex codas and avoiding onset clusters. As Bonan [1] observes, Guadeloupean (and by close analogy Martinican) Creole reduces many French CVC syllables by deleting the coda or inserting a vowel to yield CV rhythm, and most Creole words are one syllable shorter than their French counterparts (p. 72). For example, French *lavable* [lavabl](washable) surfaces in Creole as *lavab* [la.vab] via apocope (therefore erasing the final consonant cluster), while *heure* [œʁ] (hour) becomes *lè* [lɛ] with coda reduction [ʁ → ø] and initial consonant epenthesis giving us a CV sequence. Such systematic simplification and normalization directly motivates a metric of syllable count and consonant-cluster weight per token.

From an information-theoretic standpoint, we quantify character-sequence unpredictability via Shannon entropy, forming our `entropy` feature. Shannon's original formulation defines entropy as the expected information per symbol, and subsequent work has shown that languages with reduced phonotactic constraints tend to exhibit higher local entropy than their lexifiers [2,10]. In Creole orthography, this often manifests in short, morphologically stable function words like *an*, *pa*, or *sé*, whose limited character options yield low entropy, contrasting with longer, morphologically variable French forms such as *ajouter* or *régulièrement*, whose richer orthotactic patterns raise entropy. By measuring entropy at the token level after normalizing case and punctuation, we ensure the feature captures orthographic–phonotactic complexity rather than stylistic variation.

Finally, Creoles exhibit systematically different consonant–to–vowel balances, which we encode in our `vowel_ratio` feature. Large-scale phonological surveys report that Creoles cluster around roughly twenty consonants but maintain fewer vowel qualities (often seven to ten), whereas French combines a similar consonant inventory with twelve to sixteen vowel phonemes [4,5]. This difference influences the segmental makeup of words: Creole tokens such as *mango* or *pou* typically have a high vowel proportion relative to consonants, while French items like *prendre* or *structure* are consonant-heavy. This distributional cue, when combined with syllable complexity and entropy, provides a complementary dimension for distinguishing Creole from French tokens without relying on contextual information.

In our updated workflow, these three compact, interpretable features form the sole input to *both* classical and quantum pipelines. By design, we strip all contextual information, balance the Creole/Not-Creole training sets by type, and apply identical punctuation filtering to gold and predicted data. We compute features after lowercasing and canonicalizing quotes/dashes. For entropy, we retain only letters plus apostrophe and hyphen (to capture orthographic structure); for

vowel ratio and syllable complexity, we use letters only, removing punctuation and digits. Punctuation-only tokens are excluded from evaluation. This ensures that performance differences stem from the representation and classifier, not from variation in token coverage or preprocessing.

2.3 Classical vs. Quantum Pipelines

Our *classical* pipeline represents each token by a three–dimensional real-valued vector

$$\left[H_{\text{char}}, r_{\text{vowel/cons}}, \sigma_{\text{syll}}\right],$$

where H_{char} is the Shannon entropy of the character sequence, $r_{\text{vowel/cons}}$ its vowel-to-consonant ratio, and σ_{syll} a simple syllable-complexity score. After standardizing each feature with a z–score (mean and variance computed from the training data only), we train two alternative classifiers in this 3D feature space: (1) a linear logistic regression, which learns a single separating hyperplane and treats each feature's contribution additively, and (2) an RBF-kernel SVM, which uses a Gaussian kernel to model non-linear decision boundaries by implicitly mapping the data to a higher-dimensional space. In our setting, the three token-internal features are already low-dimensional, well-separated along individual axes for many tokens, and deliberately free of contextual noise. This makes them well-suited to a linear separator, while the SVM's more flexible boundary can overfit to small, spurious feature combinations in this compact space, especially under our balanced-type sampling. As a result, although SVMs can in principle capture higher-order interactions without explicit feature crossing or polynomial expansion, here this capacity offers little benefit and may even degrade generalization—motivating our choice to retain linear logistic regression as the classical baseline.

Unlike the SVM, whose nonlinear capacity resides entirely in the decision function applied to the unchanged 3D input space, the quantum circuit alters the geometry of the feature space itself before classification, embedding the three features into a higher-dimensional quantum state whose amplitudes explicitly encode joint feature products. Our *quantum* pipeline uses exactly the same three standardized features but nonlinearly embeds them into a variational quantum circuit with one qubit per feature. We evaluate two configurations:

- **Product-state circuit (no entanglement).** Each standardized feature value is embedded into a single qubit via a single R_Y rotation, producing a separable n-qubit state ($n = 3$ in our experiments). This encoding already induces a nonlinear mapping, since the expectation values of Pauli-Z measurements depend sinusoidally on the input angles.
- **Entangled-state circuit.** After the initial R_Y embeddings, we insert an *IsingXX* entangling chain with fixed angle $\pi/2$ between consecutive qubits. This operation mixes the computational subspaces of adjacent features, creating joint amplitude patterns that product states cannot represent. To further expose these correlations under Z-basis measurement, we apply a fixed $R_Y(\pi/4)$ rotation to each qubit before readout.

In both configurations, the circuit is measured via local Pauli-Z expectation values, yielding one real-valued output per qubit (three values for our three-feature setup). These outputs are passed directly to the same classical decoder as in the baseline pipeline, ensuring that any performance difference is attributable solely to the quantum embedding stage.

Our updated experimental setup uses a unified evaluation function across all models, identical gold token filtering, and balanced type-level training sets, allowing direct comparison between:

1. n-gram language-model baselines (best: 2-gram),
2. classical logistic/SVM on the 3-D feature vector,
3. product-state quantum embeddings, and
4. entangled-state quantum embeddings.

This hybrid design preserves the interpretability of the input features while allowing the quantum embedding stage to reshape the feature space through quantum interference (via RY rotations) and, when enabled, entanglement (via fixed-angle IsingXX gates followed by readout rotations). By keeping the final classification stage entirely classical, we isolate performance differences to the embedding step itself, ensuring that gains—such as the observed error reductions and improved recall for high-frequency Creole tokens—are specifically attributable to the quantum mapping of the three interpretable structural features rather than changes in the decoder architecture.

2.4 Classical Pipeline Embedding and Feature Selection

To isolate the structural distinctions between Martinican Creole and French borrowings, each token is mapped to a compact three–dimensional feature vector:

- **Character entropy (H)** measures the Shannon entropy of the token's character distribution, capturing unpredictability in orthographic form. Computation is case-insensitive and based on the empirical distribution after canonicalizing punctuation and stripping whitespace, ensuring consistency with our gold dataset preprocessing.
- **Vowel–consonant ratio (V)** computes the proportion of vowel characters in the token, exploiting the differing CV patterns in Creole vs. French.
- **Syllable complexity (S)** counts vowel–consonant transitions and cluster sizes, approximating the number and intricacy of CV units per token.

In our updated workflow, all training sets are *type-balanced* at the class level (equal number of unique Creole vs. Not–Creole tokens), and punctuation-only tokens are filtered out from both training and gold evaluation sets prior to feature computation. This ensures that support counts match across traditional, classical, and quantum experiments, eliminating discrepancies caused by pipeline-specific filtering.

These standardized features are z–scored based on the training distribution before being passed to one of two classical learners:

1. *Logistic regression* with ℓ_2 regularization.
2. *RBF-kernel SVM* with γ set to the reciprocal of the feature dimension.

While the linear model cannot capture higher-order interactions without explicit polynomial expansion, the RBF-kernel SVM can model such interactions implicitly through its kernel mapping. In our low-dimensional, well-separated feature space, however, this extra flexibility offered little benefit and sometimes degraded generalization, making logistic regression the more stable baseline.

2.5 Decoder Choice Across Pipelines

We compare two decoder architectures across three embedding settings: logistic regression (LR) with ℓ_2 regularization and an RBF-kernel SVM.

- **Classical three-feature vector.** Tokens are represented by the standardized (H, V, S) feature triplet: Shannon entropy H, vowel-to-consonant ratio V, and syllable-complexity score S.
- **Quantum product-state embedding.** Each standardized feature is encoded into a single qubit via a rotation gate, producing a separable 3-qubit state whose measurement outcomes yield nonlinear transformations of the input features.
- **Quantum entangled embedding.** As in the product-state case, but with IsingXX entanglers and readout rotations inserted between qubits, allowing the circuit amplitudes to represent higher-order correlations among H, V, and S that cannot be captured in $\mathbb{R}^3$ without explicit feature crossing.

Table 1. Logistic Regression (LR) and Support Vector Machine (SVM) performance across the classical three–feature pipeline, the non-entangled 3-qubit quantum embedding, and the entangled 3-qubit quantum embedding.

Pipeline	Decoder	$F_{1,\text{Creole}}$	$F_{1,\neg\text{Creole}}$	M$-F_1$	Acc.
Classical	LR	0.94	0.25	0.60	0.90
	SVM	0.61	0.08	0.34	0.45
Quantum (non-entangled)	LR	0.92	0.21	0.57	0.85
	SVM	0.50	0.08	0.29	0.35
Quantum (entangled)	LR	0.93	0.19	0.56	0.86
	SVM	0.60	0.08	0.34	0.44

As shown in Table 1, logistic regression consistently outperforms the RBF-kernel SVM across all pipelines, maintaining high Creole recall and reasonable performance on the minority Not-Creole class. The RBF kernel's flexibility, while powerful for high-dimensional or axis-entangled data, offers little benefit here: our (H, V, S) features are low-dimensional, interpretable, and already

well-separated along individual axes. In this compact space, the SVM is prone to overfitting to spurious combinations (particularly when the input geometry comes from the variational quantum embeddings), leading to degraded generalization.

The classical pipeline with LR achieves the highest token-level F_1, but as shown in Table 5, the entangled quantum embedding with LR surpasses all others at the type level, revealing complementary strengths between classical and quantum approaches.

2.6 1-Qubit Embedding Ablation

To evaluate the contribution of each structural feature in isolation, we trained single-qubit circuits encoding only one feature under the same logistic decoder. Table 2 reports the Creole and Not-Creole F_1 scores for each 1-qubit embedding.

Table 2. F_1 scores for Creole and Not-Creole using single-feature (1-qubit) embeddings.

Feature	$F_{1,\text{Creole}}$	$F_{1,\neg\text{Creole}}$
Syllable_Complexity (1-qubit)	0.35	0.05
Entropy (1-qubit)	0.51	0.05
Vowel_Ratio (1-qubit)	0.81	0.09

Among single-feature embeddings, *vowel ratio* achieves the strongest Creole discrimination ($F_1 = 0.81$) and the highest minority-class score ($F_1 = 0.09$). Entropy offers moderate Creole performance ($F_1 = 0.51$) but no gain on Not-Creole, while syllable complexity performs worst overall ($F_1 = 0.35$; $F_1 = 0.05$). The relatively low scores across all one-feature embeddings highlight that no single structural cue is sufficient for robust token-level separation, motivating the move to multi-feature embeddings.

2.7 2-Qubit Entanglement Effects

We next evaluated pairwise feature embeddings using both product-state (no entanglement) and entangled 2-qubit circuits for each feature pair. Table 3 reports the Creole and Not-Creole F_1 scores for all six configurations, alongside our final 3-qubit entangled result for reference.

Two trends emerge:

1. **No entanglement, no improvement over 1-qubit.** The product-state 2-qubit scores are roughly in line with the better-performing single-feature models, indicating that simply combining two features without entanglement offers little additional discrimination. In some cases, the second feature appears to *reduce* clarity, likely because unentangled qubits encode each feature independently, so any conflicting cues act more like noise than reinforcement.

2. **Entanglement consistently boosts both classes.** Adding IsingXX entangling gates ($\pi/2$ rotation) between adjacent qubits produces marked improvements for all feature pairs, especially in minority-class F_1. For example, *(entropy, vowel ratio)* jumps from $(0.84, 0.10)$ to $(0.95, 0.19)$. These gains suggest that the gates allow the circuit to represent joint amplitude correlations between features that product-state circuits cannot encode.

Table 3. F_1 scores for Creole and Not-Creole using 2-qubit product-state vs. entangled embeddings, and the full 3-qubit entangled model.

Features	Entangled?	$F_{1,\text{Creole}}$	$F_{1,\neg\text{Creole}}$
Entropy, Vowel_ Ratio	No	0.84	0.10
	Yes	0.95	0.19
Entropy, Syllable_ Complexity	No	0.67	0.09
	Yes	0.94	0.20
Vowel_ Ratio, Syllable_ Complexity	No	0.89	0.16
	Yes	0.94	0.20
All Three Features (3-qubit entangled)		**0.93**	**0.19**

While certain entangled 2-qubit pairs (e.g., *entropy, vowel ratio*) achieve very high Creole F_1, they necessarily omit one structural feature and thus cannot encode all interaction terms present in the full (H, V, S) space. The 3-qubit entangled configuration preserves every pairwise and three-way interaction in a single circuit, enabling a feature-complete comparison against classical baselines. This ensures that any observed performance differences are attributable to the *representation* (classical vs. quantum) rather than differences in feature coverage, and supports our decision to adopt the 3-feature entangled model as the primary QNLP architecture for evaluation.

2.8 Token-Level Performance at a Glance

Table 4 summarizes the end-to-end token-level results for our three final pipelines using the same (H, V, S) features and a logistic decoder. The *classical* model attains the strongest overall scores (macro-$F_1 = 0.60$, accuracy $= 0.90$), with high Creole F_1 and moderate Not-Creole F_1. The two *quantum* embeddings remain competitive: the product-state variant trades a small drop in Creole F_1 for the best minority recall (0.53), while the entangled variant slightly improves Creole F_1 over product-state but gives back some Not-Creole recall (0.44).

A look at recalls clarifies the trade-offs. Creole recall is highest for the classical model (0.91), followed by the entangled (0.88) and product-state (0.87) embeddings. For the minority class, the ranking flips: product-state achieves the best Not-Creole recall (0.53), classical is next (0.47), and entangled is lowest (0.44).

Table 4. Token-level results on the filtered 7,843-token evaluation set (punctuation removed).

Pipeline	$F_{1,\text{Creole}}$	$F_{1,\neg\text{Creole}}$	M$-F_1$	Acc.
Classical (3 features, LR)	0.94	0.25	0.60	0.90
Quantum (3-qubit, product-state, LR)	0.92	0.21	0.57	0.85
Quantum (3-qubit, entangled, LR)	0.93	0.19	0.56	0.86

Given the skewed gold distribution (≈96% Creole), these shifts in recall dominate macro-F_1, explaining why the classical model leads in aggregate despite quantum embeddings capturing non-linear feature interactions.

Crucially, token-level metrics conflate frequency with generalization. In Sect. 3, we show that *type-level* evaluation tells a different story: the entangled embedding substantially reduces the number of distinct lexical types that are ever misclassified (284 vs. 404 for classical and 509 for product-state), and it uniquely fixes large swaths of types that both classical and product-state models miss. This motivates analyzing performance through the lens of types, where robustness to unseen or infrequent forms matters most in mixed-language settings.

3 Type-Level Error Analysis

Token-level scores provide a coarse view of model behavior, but obscure the fact that many errors are concentrated on a small subset of lexical types that recur frequently in the corpus. To better understand systematic strengths and weaknesses, we perform a *type-level* error analysis.

3.1 Methodology

Starting from the 7,843-token test set, we collapse tokens into unique types after normalizing for case and diacritics. This yields N_T distinct types, each associated with a gold label (*Creole* or *Not-Creole*). When an identical string appears in both datasets but with different labels (e.g., French in one, Creole in the other), we treat these as distinct homophonous types, each with its own label, rather than merging them into a single type.

For each pipeline (Classical, Quantum product-state, Quantum entangled), we then determine:

1. Types *correctly* classified by all models.
2. Types *misclassified* by all models (*common errors*).
3. Types misclassified by only one model (*exclusive errors*).
4. Pairwise overlaps of misclassified types between models.

We report counts for each category. This approach reveals whether gains in token-level F_1 translate into broader lexical coverage or merely reclassify high-frequency tokens.

3.2 Exclusive Vs. Common Errors

Table 5 summarizes the total number of misclassified types for each model and the exclusive overlaps between error sets.

Table 5. Type-level misclassification counts and exclusive overlaps between error sets for the Classical, Quantum Product-state, and Quantum Entangled pipelines.

Error set	# Types
Classical only	0
Product-state only	145
Entangled only	78
Common to all three	166
Classical ∩ Product only (not Entangled)	198
Classical ∩ Entangled only (not Product)	40
Product ∩ Entangled only (not Classical)	0
Total type errors—Classical	404
Total type errors—Product-state	509
Total type errors—Entangled	**284**
Total types - Creole	1762
Total types - Not Creole	173
Total types	1 935

Several patterns are immediately noteworthy:

- **Entangled has the fewest total type errors** (284), far below Classical (404) and especially Product-state (509). This implies that quantum embeddings are not uniformly beneficial. On the contrary, mapping the three features into a quantum feature space without entanglement (product-state) can *harm* type-level accuracy compared to the Classical baseline, indicating that quantum representation alone is not sufficient; the observed gains appear only when entanglement is introduced, an effect that is only possible within the quantum feature space—at which point we observe not just error correction but the strongest overall type-level performance. This highlights that quantum circuits can yield both improvements and degradations: they can degrade performance if used without feature entanglement, yet deliver clear gains when entanglement is leveraged effectively.
- **Classical has no unique type errors**, meaning every type it misclassifies is also misclassified by at least one quantum model. This indicates a complete overlap between the Classical error set and the union of quantum error sets: the quantum pipelines, whether entangled or not, never recover a type that the Classical model alone gets wrong. Instead, they either match the Classical

errors or introduce new ones of their own, depending on the embedding configuration. This suggests that the representational shift into quantum space fundamentally changes decision boundaries rather than refining the Classical model's existing strengths.
- The largest exclusive overlap is **Classical ∩ Product only** (198 types): types that both non-entangled models get wrong but the Entangled model gets right. This is a strong indicator of the added representational benefit of feature entanglement, which seems to resolve specific lexical clusters missed by both simpler baselines.
- **Product-state only** errors are numerous (145 types), consistent with the idea that without entanglement, the quantum model overgeneralizes structural cues, producing many avoidable misclassifications.
- **Entangled only** errors (78 types) are relatively rare and represent the smallest exclusive set among the three pipelines, reinforcing its overall robustness.
- There are **no types in the Product ∩ Entangled only** set. When both quantum variants misclassify a type, the Classical model is always wrong as well. This suggests that certain types are systematically challenging regardless of representation.

Even before a detailed qualitative error analysis, these counts reveal that type-level error reduction is the most visible gain from entanglement. While token-level F_1 scores in Table 4 show only modest improvements over the Classical baseline, the Entangled model demonstrates markedly better lexical coverage, recovering entire sets of types that remain problematic for both the Classical and Product-state pipelines. This suggests that in real-world deployment, where models regularly encounter unseen types, entanglement could yield more reliable generalization than token-level metrics alone might imply.

3.3 Errors Introduced by the Product-State Model

The **145 types** misclassified exclusively by the product-state quantum pipeline represent **7.49%** of the total 1 935 types in the dataset. Breaking this down by gold label:

- **Creole types:** 138/1762 (**7.83%** of all Creole types)
- **Non-Creole types:** 7/173 (**4.05%** of all non-Creole types)

While the percentage is smaller for non-Creole types, the two groups reveal distinct error tendencies:

- **Creole types** are dominated by proper nouns (place names, personal names) and foreign-origin items adapted to Creole orthography (e.g. *Chelchè*, *Jézikri*, *Jamayik*, *Brital*). These often contain digraphs, accented vowels, or consonant clusters atypical of French, yet still share partial orthographic cues with it, potentially misleading the product-state model's decision boundary.

– **Non-Creole types** are mostly standard French lexical items or compounds (*générations*, *surveillance*) and semantically marked borrowings (*urne/ballot*, *fossoyeur/gravedigger*), some with morphological complexity or orthographic patterns absent from Creole.

This suggests that the **absence of entanglement** leaves the product-state circuit vulnerable to:

1. Over-sensitivity to surface orthographic similarity with the opposite class.
2. Misinterpretation of rare or morphologically complex forms that deviate from the dominant class's core distribution.

The fact that most of these tokens are correctly classified by the entangled model strengthens the interpretation that entanglement helps integrate feature interactions in a way that disambiguates rare or orthographically deceptive types.

4 Effect of Feature Entanglement at the Type Level

At the type level, the clearest advantage of entanglement appears when examining types that both the classical and product-state pipelines misclassify but that the entangled model predicts correctly. The **198 types** misclassified by the product-state and classical models but not by the entangled model represent 10.23% of the total 1 935 types in the dataset. Breaking this down by gold label:

– **Creole types:** 185/1762 (10.5% of all Creole types)
– **Not-Creole types:** 13/173 (7.51% of all non-Creole types)

Table 6 presents representative examples from this exclusive intersection. All items in this set were consistently misclassified by the first two pipelines, yet correctly tagged by the entangled model.

These types include high-frequency function words, polyfunctional lexical items, compounds, proper names, and morphologically complex forms. Several are proper names that have not been orthographically adapted into Creole and/or appear with identical spelling across multiple languages. Others bear orthographic cues associated with French (e.g. capitalization, accented vowels, French-like consonant clusters) yet are labeled as Creole in the gold annotation due to their sentence-level context. It is important to note that *our models do not have access to this context*—classification is based solely on three intra-token features (entropy, vowel ratio, syllable complexity). Consequently, the fact that the entangled model "rescues" these cases is a fortunate coincidence: the same token could just as easily have been misclassified as Not-Creole had it appeared in the dataset as a Not-Creole type. From a methodological standpoint, this is expected in our proof-of-concept setup, where the aim is to test whether feature entanglement can yield measurable gains, not to produce context-aware, state-of-the-art LID.

Table 6. Representative types misclassified by both the classical and product-state models but correctly tagged under entanglement.

Token	Gold Class	Example Type
y	Creole	High-freq. function word
w	Creole	High-freq. function word
toubannman	Creole	Lexical adverb
bagay	Creole	Polyfunctional noun
vréyé	Creole	Verb (French-like orthography)
bannann	Creole	Common noun
manmay	Creole	Common noun
LEBIELLE	Creole	Proper name
BERNABE	Creole	Proper name
jwenn	Creole	Verb
mai	Not_Creole	Month name
président	Not_Creole	Loanword, accented vowel
en	Not_Creole	French preposition
Mai	Not_Creole	Month name, capitalized
OU	Not_Creole	Uppercase French coord. conj.
QUATRE-VINGT	Not_Creole	French numeral
Voie	Not_Creole	French noun
aussi/too	Not_Creole	Code-mixed token
autorités	Not_Creole	French plural noun
die	Not_Creole	English translation

4.1 Residual Classical–Entangled Overlap

A small set of types is misclassified by both the classical and entangled pipelines, but correctly classified by the product-state model, indicating cases where feature entanglement does not confer an advantage over the classical baseline.

The **40 types** in this overlap represent **2.07%** of the total 1 935 types in the test set. Breaking this down by gold label:

- **Creole types:** 22/1762 (**1.25%** of all Creole types)
- **Not-Creole types:** 18/173 (**1.02%** of all Not-Creole types) (Table 7)

Several patterns emerge:

- These residual errors are rare overall, suggesting that the classical and entangled models typically diverge in their mistakes.
- Most are **Creole types** with morphological complexity or compounding (e.g. `djoubak-tala`, `linivèsité-a`, `vakabonnajri-tala`), which may yield ambiguous intra-token statistics.

Table 7. Representative types misclassified by both the classical and entangled pipelines but correctly classified by the product-state model.

Token	Gold Class	Example Type
djoubak-tala	Creole	Noun + demonstrative
linivèsité-a	Creole	Definite NP (noun + article)
sébwet-lin	Creole	Hyphenated compound noun
vakabonnajri-tala	Creole	Noun + demonstrative
pwogré-tala	Creole	Noun + demonstrative
etc.	Not_Creole	Abbreviation / internationalism
commune	Not_Creole	French noun (municipality)
Efel	Not_Creole	Proper name
Saint-Lucia	Not_Creole	Country name
Fête	Not_Creole	French common noun

- The **Not-Creole types** often include proper names, place names, and French nouns whose orthography is shared across languages (e.g. `Saint-Lucia`, `commune`, `Fête`).
- Abbreviations and internationalisms (`etc.`, `Efel`) remain challenging due to their orthographic neutrality.

4.2 New Type Errors Introduced by Entanglement

A limited set of types is misclassified only by the entangled model, indicating cases where feature entanglement shifts decision boundaries in disadvantageous ways.

Table 8. Representative types correctly classified by the product-state model but misclassified by the entangled variant.

Token	Gold Class	Example Type
1é	Creole	Numeric form with diacritic
ou	Creole	High-freq. function word
osi	Creole	Polyfunctional adverb
o	Creole	Short function word
lè-a	Creole	Definite NP (article + noun)
lespri-yo	Creole	Noun + plural marker
lutter	Not_Creole	French verb (infinitive)
officer	Not_Creole	English translation
nouvelle	Not_Creole	French feminine adjective/noun
nom/name	Not_Creole	Code-mixed form

The **78 types** misclassified exclusively by the entangled pipeline represent **4.03%** of the total 1 935 types in the test set. Breaking this down by gold label:

- **Creole types:** 56/1 762 (**3.18%** of all Creole types)
- **Not-Creole types:** 22/173 (**12.72%** of all Not-Creole types) (Table 8)

Several patterns emerge:

- These entangled-only errors are rare overall, showing that entanglement does not typically introduce large numbers of new mistakes.
- Most are **Creole types** with multi-morphemic or cliticised forms (e.g. `lè-a`, `lespri-yo`) or orthographic irregularities (`1é`), where feature interactions may yield unexpected similarity scores.
- A smaller subset are **French or English borrowings** (e.g. `lutter`, `officer`, `nouvelle`) whose internal statistics align more closely with Creole types under the entangled encoding.
- Many are short function words or hybrids (`o`, `ou`, `nom/name`) that sit near decision boundaries in the feature space.

From a linguistic perspective, these cases share a common property of *internal heterogeneity* (i.e., mixed orthography, morphological complexity, or inflected borrowings) which is not fully captured by our current feature set. Since our objective here is to evaluate whether entanglement offers a measurable benefit in a minimal, interpretable setting, such errors are not unexpected. A next step for context-aware QNLP would be to incorporate subword morphological embeddings or sentence-level features to address these limitations.

5 Discussion

Our pipeline deliberately frames each Creole vs. Not-Creole decision as an isolated, token-level classification task, without access to sentence- or document-level context. This choice was motivated by two constraints: (1) maintaining quantum circuits compact enough for deployment on NISQ-era hardware, and (2) isolating the effect of three interpretable intra-token features—entropy, vowel ratio, and syllable complexity—when embedded in product-state and entangled configurations.

This design enables clear attribution of gains or losses to the quantum embedding itself, but it also leaves certain error clusters unresolved. In particular, short, high-frequency function words shared across languages remain difficult to disambiguate without contextual cues. In a practical language-identification system, token-level predictions from our quantum model could be combined with sequence-level smoothing or hybrid classical–quantum architectures to incorporate surrounding context.

Restricting the embedding to three features allowed a direct mapping from each feature to a single-qubit rotation, keeping circuit depth minimal, but inevitably limited representational richness. Type-level error patterns suggest

that morphologically complex words, longer compounds, and orthographically French-like borrowings could benefit from additional descriptors such as subword frequency profiles, morphological segmentation, or stress-pattern cues. Incorporating these while preserving interpretability is a natural next step.

Future Directions

Building on these findings, we see four immediate avenues for extension:

- **Context-aware QNLP:** Embed the current quantum token representation in hybrid sequence models to combine intra- and inter-token cues.
- **Expanded feature sets:** Incorporate additional linguistically interpretable features—morphological, prosodic, and distributional—while balancing circuit complexity with NISQ constraints.
- **Entanglement strategies:** Systematically compare different multi-qubit entangling patterns to determine whether the gains observed here generalize across correlation structures.
- **Cross-lingual transfer:** Evaluate portability to other Creole and closely related language pairs, including low-resource code-switching scenarios.

Overall, combining compact, interpretable quantum embeddings with carefully designed entanglement yields measurable type-level gains in low-resource language identification. This proof of concept illustrates how QNLP can bridge theoretical quantum advantages and concrete linguistic utility, especially in settings where context is limited but token-internal cues are strong.

6 Conclusion

We presented the first QNLP proof of concept for token-level language identification in Martinican Creole, comparing a character-bigram baseline, a classical logistic model on three interpretable structural features (entropy, vowel ratio, syllable complexity), and a compact three-qubit variational circuit in product-state and entangled configurations.

While token-level metrics show the classical model slightly ahead of the entangled variant, the type-level analysis reveals the opposite: entanglement reduces the number of distinct error types, rescuing items consistently misclassified by both classical and product-state pipelines. These rescued types are heterogeneous (frequent Creole items, compounds, and proper names with cross-lingual orthography).

The small set of errors unique to the entangled model (4.03% of types) often involves internally varied forms whose feature profiles trigger over-sensitivity to spurious correlations. A residual overlap with the classical model (2.07% of types) points to persistent cross-lingual orthographic ambiguity and complex derived forms.

These findings demonstrate that even minimal quantum embeddings can capture non-linear intra-token correlations beyond the reach of n-gram baselines

and simple linear models, yielding measurable type-level gains in a low-resource, hardware-friendly setting. Such gains support the integration of entanglement into future low-resource LID pipelines—especially where context is scarce but token-internal cues are strong.

References

1. Bonan, C.: The Core Grammar of Guadeloupean Creole. A Descriptive and Comparative Approach. Università Ca'Foscari Venezia (2013)
2. Cover, T.M., Thomas, J.A., et al.: Entropy, relative entropy and mutual information. Elem. Inf. Theor. **2**(1), 12–13 (1991)
3. Jaech, A., Mulcaire, G., Ostendorf, M., Smith, N.A.: A neural model for language identification in code-switched tweets. In: Proceedings of the Second Workshop on Computational Approaches to Code Switching, pp. 60–64 (2016)
4. Maddieson, I.: Consonant inventories. The World Atlas of Language Structures Online (2013)
5. Maddieson, I.: Vowel quality inventories. In: The World Atlas of Language Structures Online (2013)
6. Molina, G., et al.: Overview for the second shared task on language identification in code-switched data. arXiv preprint arXiv:1909.13016 (2019)
7. Myers-Scotton, C.: Constructing the frame in intrasentential codeswitching. Multilingua **11**(1), 101–127 (1992)
8. Poplack, S.: "sometimes i'll start a sentence in spanish y termino en español": toward a typology of code-switching. Linguistics **51**(s1), 11–14 (2013)
9. Schramm, M.: The Emergence of Creole Syllable Structure: A Cross-Linguistic Study, vol. 554. Walter de Gruyter GmbH & Co. KG (2014)
10. Shannon, C.E.: A mathematical theory of communication. Bell Syst. Tech. J. **27**(3), 379–423 (1948)